

# Variance of OLS Estimators and Residual Variance Estimation

## 1 The Story Behind Linear Regression and Estimation

Linear regression models the relationship between a dependent variable  $y$  and independent variables  $X$  as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (1)$$

where:

- $\mathbf{y} \in \mathbb{R}^n$  is the vector of observed values.
- $\mathbf{X} \in \mathbb{R}^{n \times p}$  is the design matrix.
- $\boldsymbol{\beta} \in \mathbb{R}^p$  is the vector of unknown coefficients.
- $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$  is a vector of i.i.d. errors.

The errors  $\boldsymbol{\varepsilon}$  satisfy:

- Zero mean:  $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ .
- Constant variance:  $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ .
- Independence: Errors  $\varepsilon_i$  are i.i.d.

Understanding these assumptions is crucial as they allow us to derive important properties about our estimators, including their variance.

## 2 Estimating $\boldsymbol{\beta}$ Using OLS

The ordinary least squares (OLS) estimator minimizes the sum of squared errors. This means that we seek a coefficient vector  $\boldsymbol{\beta}$  that makes our predictions as close as possible to the actual observed values:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2. \quad (2)$$

Solving this equation gives:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (3)$$

This estimator provides the best linear unbiased estimate of  $\boldsymbol{\beta}$ , meaning it has the lowest variance among all unbiased linear estimators, assuming our assumptions hold.

### 3 Variance of $\hat{\beta}$

To compute the variance of  $\hat{\beta}$ , substitute  $\mathbf{y}$  into the estimator:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X} \beta + \varepsilon). \quad (4)$$

Expanding:

$$\hat{\beta} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \varepsilon. \quad (5)$$

This equation shows that our estimate  $\hat{\beta}$  consists of two parts: the true  $\beta$  and an error term that depends on  $\varepsilon$ . Since  $\varepsilon$  is random, so is  $\hat{\beta}$ , leading us to investigate its variance.

Taking the variance:

$$\text{Var}(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \text{Var}(\varepsilon) \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}. \quad (6)$$

Since  $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}$ , this simplifies to:

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}. \quad (7)$$

This result tells us how the uncertainty in our estimates depends on the spread of the independent variables in  $\mathbf{X}$ .

### 4 Estimating $\sigma^2$ Using Residuals

Since  $\sigma^2$  is unknown, we estimate it using residuals. Residuals measure how much each observed value deviates from the predicted value.

#### 4.1 Compute Predicted Values

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\beta}. \quad (8)$$

#### 4.2 Compute Residuals

$$\hat{\varepsilon} = \mathbf{y} - \mathbf{X} \hat{\beta}. \quad (9)$$

#### 4.3 Compute Residual Sum of Squares (RSS)

$$\text{RSS} = (\mathbf{y} - \mathbf{X} \hat{\beta})^T (\mathbf{y} - \mathbf{X} \hat{\beta}). \quad (10)$$

#### 4.4 Estimate $\sigma^2$

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n - p}. \quad (11)$$

This correction by  $n - p$  instead of  $n$  ensures an unbiased estimate of  $\sigma^2$ , accounting for the degrees of freedom used in estimating  $\beta$ .